WENTWORTH LIU

Beijing, China · Eligible to work in the UK without sponsorship · Open to relocate to London
gzliu0705@gmail.com · linkedin.com/in/guanzhang-liu-4a4a02413
Data Scientist · User Behaviour Analytics · Predictive Modelling · Targeting & Prioritisation

SUMMARY

Data Scientist with 2+ years of experience taking ambiguous, high-value business questions from 0–1: framing the decision, building behavioural metrics and data foundations, developing explainable models, and embedding outputs in cross-functional workflows. Applied this end to end to million-scale driving and charging behaviour at Li Auto, and to user segmentation and content targeting at ByteDance.

EXPERIENCE

Li Auto Beijing, China
Data Scientist, Battery Intelligence Department Jul 2024 – Present

- **0–1 Behavioural Risk Targeting for Battery Safety.** Faced tens of thermal-runaway cases among 100k–700k vehicles, where a binary classifier would be statistically weak and operationally unhelpful. Defined a cross-platform framework of driving, charging, battery-state, and lifecycle signals; reframed the task as explainable relative-risk ranking to identify vehicles with failure-like behaviour patterns. Built and deployed a monthly pipeline that captured 93% of newly observed failures in the top 2% of ranked vehicles, creating an actionable investigation priority queue for quality and after-sales teams.
- **0–1 User-Behaviour Decision System for Battery Longevity.** Broke down the business question from "which behaviours correlate with degradation?" to "which behaviours are actionable without compromising user experience?" Built cohort definitions, quality-controlled SOH outcomes, and causal analysis using propensity-score matching and treatment-effect estimation to separate charging-temperature and SOC effects from confounders. Partnered with BMS, cell-design, performance, and after-sales teams to translate findings into OTA/BMS strategy design and battery-design boundary tests; scenario analysis projected tens of millions of RMB in warranty-exposure reduction for one model.
- **End-to-End Data and Model Foundation.** Replaced fragmented, notebook-specific behaviour features with a team-wide, configuration-driven offline feature store: standardised VIN-cycle metrics, feature definitions, data lineage, PySpark transformations, and train–score consistency across five battery platforms. Made behavioural analysis and model outputs reproducible and reusable for the group, reducing feature-development time by ~85%.
- **Model-to-Decision Workflow Adoption.** Designed and AI-assisted-developed an internal risk-analysis workbench so battery and after-sales analysts could compare cohorts, investigate individual vehicles, interpret behavioural drivers, and benchmark against historical failures. Turned model scores from a technical output into an auditable workflow for investigation, feedback, and root-cause analysis.

ByteDance Beijing, China
Data Analytics Intern, Douyin Market Strategy May – Sep 2023

- **User Segmentation, Targeting & Content Prioritisation.** Built a 0–1 measurement framework combining consumption, dissemination, and search signals from 5+ data platforms; redesigned content metrics and taxonomy for Qishui Music. Diagnosed new-user conversion decline through funnel segmentation and KOL audience-saturation analysis, then converted insights into content and campaign prioritisation; targeted campaigns validated ~40% consumption growth. Added monitoring and automation to make the workflow dependable for recurring decisions.

Dongwu Securities China
Equity Research Intern, Mechanical Sector Dec 2021 – Mar 2022

- **Decision-focused Quantitative Analysis.** Combined company fundamentals, competitive positioning, and market dynamics with stock-price modelling to support investment recommendations for Harmonic Drive Systems.

EDUCATION

Nanjing University Sep 2021 – Jun 2024
MSc Computational Materials Science · First-author, *Physical Review B* (2024); co-author, *Physical Review B* (2024) and *Physical Review Materials* (2023); Patent: CN115795266A.

Nanjing University Sep 2017 – Jul 2021
BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort · National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention.

SKILLS

Programming	Python · SQL/HiveSQL · PySpark · C++ · MATLAB
Data Science	Predictive Modelling · Statistical Modelling · Causal Inference / PSM · User Segmentation · Feature Engineering · NLP · Model Evaluation · LightGBM · scikit-learn
Delivery	Targeting & Prioritisation · Behavioural Metrics · Batch ML Pipelines · Data Quality · Monitoring · Feature Store · Explainability · Decision Support
Languages	English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)